ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PRACTICE EXAM 6 (PAPER 2) · MARKING GUIDELINE

MARKS: 80

Marking guideline for the Academic Central NC(V) Level 2 Electronics and Digital Concepts for Robotics practice exam 6 (paper 2).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Circuits and breadboard layouts may look different and still be correct: check that each connection is made. Deduct 1 mark for each wrong connection and 2 marks for any short circuit (no less than 0 for a question).

QUESTION 1

1.1 Battery with correct polarity ✓✓; diode in series with the supply ✓✓, facing the correct way (anode towards battery +) ✓✓; push button in series ✓✓; two LEDs ✓✓ each with its own 330 Ω resistor in series ✓✓✓✓; the two LED branches in parallel ✓✓; LED polarity correct ✓✓; return to battery − and neat layout / labels ✓✓ (20)

1.2 Placement, 1 mark each ✓✓✓✓✓✓✓✓ (battery, S1, R1, C1, R2, Q1, R3, LED) = 8. Polarity ✓✓✓: C1 + towards the R1 / R2 junction; LED anode towards R3; transistor pins C-B-E correct = 3. Connections ✓✓✓✓✓✓✓✓✓: battery to the rails; S1 from +9 V to R1; R1 to the C1 / R2 junction; C1 − to 0 V; R2 to the base; R3 from +9 V to the LED anode; LED cathode to the collector; emitter to 0 V; no short circuits = 9. (20)

[40]

QUESTION 2

2.1 Placement, 1 mark each ✓✓✓✓✓✓✓✓ (battery, S1, R1, Q1, relay, D1, R2, LED2) = 8. Polarity ✓✓✓: diode cathode to +9 V; LED2 anode towards R2; transistor pins = 3. Connections ✓✓✓✓✓✓✓✓✓: S1 from +9 V to R1; R1 to the base; emitter to 0 V; relay coil from +9 V to the collector; D1 across the coil; relay COM to +9 V; relay NO to R2; R2 to LED2; LED2 cathode to 0 V = 9. (20)

2.2 Correct circuit diagram: 9 V battery ✓✓ → switch S1 ✓✓ → potentiometer used as a variable resistor (one end and wiper) ✓✓✓ → junction point ✓✓; from the junction: 220 Ω resistor ✓✓ in series with the LED ✓✓ (correct direction ✓) to the − line; buzzer ✓✓ from the same junction to the − line (in parallel with the resistor-LED branch) ✓✓; correct symbols and labels throughout ✓✓ (20)

[40]

TOTAL: 80

END OF MARKING GUIDELINE